The Dehn Test Picks the Lone Tilers

Sweeps and Truncated Tetrahedra That Fill Space

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DICTO proposed filling the recorded gaps of the Concordance with a combination of new and current methods; these results fill the Dehn column of its grid of tests and families and the row of the tetrahedron sweep (The DICTO Concordance, DICTO DOI: 10.5281/zenodo.23934363). Computation, proofs, figures and drafting with Claude (Anthropic).

Version 2 corrects the notes on methods: the names of our computers and private file paths are removed, with small wording fixes. No result changes.

Some solids fill space on their own, like bricks; most do not. In 1901 Max Dehn found a ledger that every lone space-filler must balance: each edge enters with its length, written against its angle, and angles that are whole fractions of a half-turn count as nothing. Run along a family of solids, the ledger balances only at a few members. On the families studied here, those few are exactly the members that fill space, once a second, local count of angles round an edge is added. And among tetrahedra with their four corners cut off, the ledger balances exactly when the four cuts add up to one edge: every such piece we could test fills space, two to a period.

Abstract. A tetrahedron with its four corners cut off by fractions c_1, …, c_4 of the edge has Dehn invariant $6a(1 - \Sigma c_i) \otimes \arccos(1/3)$ (**Proposition 2.1**); so it can fill space alone only when the cuts add up to one edge, and that is the same as cutting the tetrahedron T by the inverted tetrahedron of twice its size, 2(−T), placed anywhere (**Proposition 2.2**). For cuts by whole layers of the octet it is also the same as keeping the octet's ratio of two tetrahedra to every octahedron (**Proposition 2.3**). **Theorem 2.4** (computer-checked, exact cover): all twelve Dehn-zero integer truncations with $n \le 5$, and (5, 3, 0, 0) at $n = 8$, tile space, each with two pieces per period, the piece and its point inverse; we conjecture this for every rational cut (**Conjecture 2.5**) and list the seven golden cut vectors that are the next to try (**Proposition 2.6**). On the sweeps the same test picks the lone tilers: on the cube–octahedron sweep the Dehn invariant vanishes exactly at Kelvin's truncated octahedron and the cube (**Theorem 1.1**); on the CO–RD sweep it vanishes on the stretch $1 \le \mu \le 4/3$ and at the B3 permutohedron, and the angle count leaves only the rhombic dodecahedron (**Theorem 1.2**); on the tetrahedron sweep it vanishes only at $\mu = 2$, the known β-tin tiler (**Proposition 1.3**).

1. The test

For a polyhedron P with edges e of length L_e and interior dihedral angle θ_e, the **Dehn invariant** is $D(P) = \Sigma_e L_e \otimes \theta_e$ in $\mathbb{R} \otimes_{\mathbb{Q}} (\mathbb{R}/\mathbb{Q}\pi)$ (Dehn 1901). It is additive when solids are glued, and it is zero for every polyhedron that tiles space by congruent copies (Debrunner 1980); by Sydler (1965) $D(P) = 0$ is also exactly the condition for P to be cut into finitely many

pieces that reassemble into a cube. Two facts are used throughout. If θ is not a rational multiple of π, then $L \otimes \theta = 0$ only for $L = 0$; and $a \otimes \theta + b \otimes \theta' = 0$ with a, b linearly independent over $\mathbb{Q}$ forces $\theta \equiv \theta' \equiv 0$ modulo $\mathbb{Q}\pi$.

The angles met here are $\beta = \arccos(1/\sqrt{3}) \approx 54.74°$ and its relatives $\alpha = \arccos(1/3) = \pi - 2\beta \approx 70.53°$, the tetrahedron's dihedral. Neither is a rational multiple of π: if $\theta = r\pi$ with r rational, $2\cos\theta$ is an algebraic integer, but $2\cos\alpha = 2/3$ and $2\cos 2\beta = -2/3$ are not (Conway, Radin and Sadun 1999 classify all such angles).

The test is one-sided: $D(P) = 0$ does not make P a tiler. The second test is local. Round a generic point of an edge of a tile, the tiles that meet there fill a full turn, so the dihedral angles of the edges through that point, plus π for every tile in whose face the point lies, add up to 2π; on the boundary of a half-space they add up to π. If no sum of the solid's own dihedral angles and multiples of π reaches 2π at some edge, the solid cannot tile alone. We call this the **angle count**.

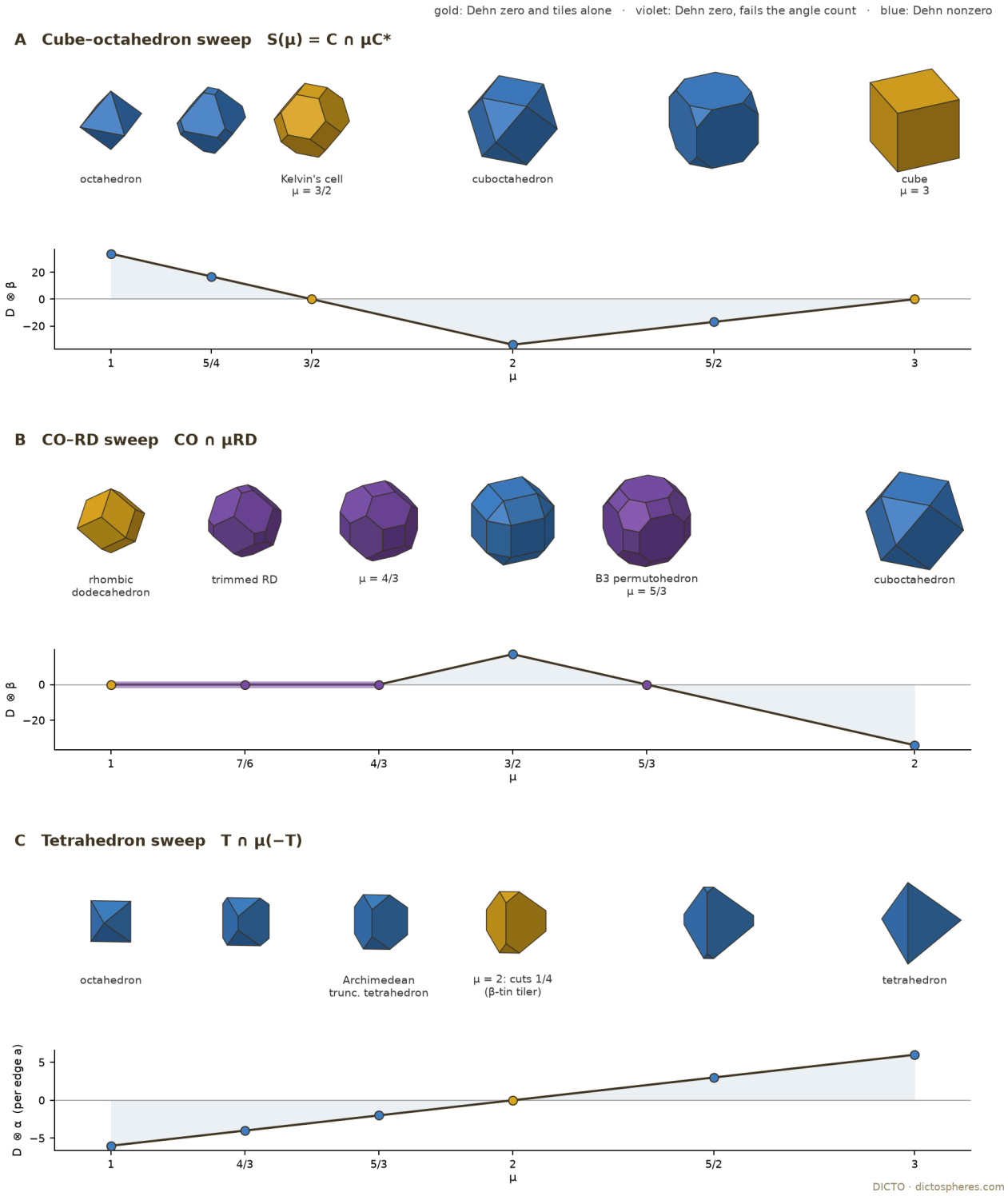


Figure 1. The three sweeps and their exact Dehn lines (Theorems 1.1, 1.2 and Proposition 1.3). Each row shows members of the sweep at the values of μ marked on the axis below, and the coefficient of the Dehn invariant against β (rows A and B, cube of side 2) or against α per unit edge (row C). Gold: Dehn zero and tiles space alone (the octahedron–cube sweep's Kelvin cell and cube; the rhombic dodecahedron; the β-tin piece at μ = 2). Violet: Dehn zero but ruled out by the angle count (the cube-trimmed rhombic dodecahedra $1 < \mu \le 4/3$, drawn at 7/6 and 4/3, and the B3 permutohedron at 5/3). Blue: Dehn nonzero. The render script recomputes the Dehn coefficient of every drawn solid from its hull and checks it against the formulas.

2. The cube–octahedron sweep

The sweep of #90 and #94 is $S(\mu) = C \cap \mu C^*$, with $C = [-1, 1]^3$ the cube and $C^* = \{|x| + |y| + |z| \le 1\}$ its polar octahedron, from the octahedron (μ = 1) through Kelvin's truncated

octahedron (μ = 3/2) and the cuboctahedron (μ = 2) to the cube (μ = 3) (#90, DICTO DOI: 10.5281/zenodo.23931136; #94, DICTO DOI: 10.5281/zenodo.23934349).

> **Theorem 1.1 (cube–octahedron sweep).** $D(S(\mu)) = 24\sqrt{2}\,(3 - 2\mu) \otimes \beta$ for $1 \le \mu \le 2$, and $D(S(\mu)) = -24\sqrt{2}\,(3 - \mu) \otimes \beta$ for $2 \le \mu \le 3$. Hence $D(S(\mu)) = 0$ exactly at $\mu = 3/2$, Kelvin's truncated octahedron, and at $\mu = 3$, the cube: the two members of the sweep that tile space alone.

Proof. For $1 \le \mu \le 2$ the faces are six squares $x = \pm 1$, $\pm y$, $\pm z$ (side $(\mu - 1)\sqrt{2}$) and eight hexagons in the planes $\pm x \pm y \pm z = \mu$. A hexagon–hexagon edge, e.g. from $(1, \mu - 1, 0)$ to $(\mu - 1, 1, 0)$, has length $\sqrt{2}\,(2 - \mu)$, and there are 12; its dihedral is $\pi - \alpha = 2\beta$. A square–hexagon edge has length $\sqrt{2}\,(\mu - 1)$, there are 24, and its dihedral is $\pi - \beta$. So $D = 12\sqrt{2}\,(2 - \mu) \otimes 2\beta - 24\sqrt{2}\,(\mu - 1) \otimes \beta = 24\sqrt{2}\,(3 - 2\mu) \otimes \beta$. For $2 \le \mu \le 3$ the faces are six octagons on $x = \pm 1$, … and eight triangles with corners $(1, 1, \mu - 2)$ and its permutations, side $\sqrt{2}\,(3 - \mu)$; octagon–octagon edges have dihedral $\pi/2$ and count as nothing, the 24 triangle–octagon edges have dihedral $\pi - \beta$, so $D = -24\sqrt{2}\,(3 - \mu) \otimes \beta$. The two formulas agree at the cuboctahedron, and at $\mu = 1$ the first gives the octahedron's own $12\sqrt{2} \otimes 2\beta$. Since $\beta \notin \mathbb{Q}\pi$, $D = 0$ only where the coefficient vanishes. Kelvin's cell and the cube both tile space. ∎

3. The CO–RD sweep

The second dual sweep of #94 is $CO \cap \mu RD$, with $CO = \{|x_i| \le 1, |x| + |y| + |z| \le 2\}$ the cuboctahedron and $RD = \{|x_i| + |x_j| \le 1\}$ its polar rhombic dodecahedron, from the RD ($\mu = 1$) to the CO ($\mu = 2$).

> **Theorem 1.2 (CO–RD sweep).** (a) $D(CO \cap \mu RD) = 0$ for $1 \le \mu \le 4/3$; $D = 72\sqrt{2}\,(\mu - 4/3) \otimes \beta$ for $4/3 \le \mu \le 3/2$; $D = 24\sqrt{2}\,(5 - 3\mu) \otimes \beta$ for $3/2 \le \mu \le 2$. So the Dehn-zero members are the stretch $1 \le \mu \le 4/3$ and the single solid at $\mu = 5/3$.
> (b) The member at $\mu = 5/3$ is, scaled by 3, the **B3 permutohedron**: its 48 corners are all signed permutations of (1, 2, 3); it has 6 octagons, 8 hexagons and 12 rectangles, and edges $\sqrt{2}$ and 2 (it is not the Archimedean truncated cuboctahedron).
> (c) Of all the Dehn-zero members, only the RD ($\mu = 1$) tiles space alone. The cube-trimmed RDs ($1 < \mu \le 4/3$) and the B3 permutohedron fail the angle count.

Proof. (a) For $\mu \le 4/3$ the RD's three-fold corners $(\mu/2, \mu/2, \mu/2)$ satisfy $|x| + |y| + |z| \le 2$, so no CO plane is active; the faces lie in the RD planes and, for $\mu > 1$, in the cube planes $x = \pm 1$. The dihedral angles are $2\pi/3$ (RD plane against RD plane) and $3\pi/4$ (RD plane against cube plane), both rational multiples of π, so $D = 0$. For $\mu > 4/3$ triangles appear in the planes $\pm x \pm y \pm z = 2$. Their dihedral is $\pi/2 + \beta$ against an RD plane and $\pi - \beta$ against a cube plane, so $D = (L_{tr} - L_{ts}) \otimes \beta$, where L_{tr}, L_{ts} are the total lengths of the triangle–RD and triangle–cube edges. For $4/3 \le \mu \le 3/2$ a triangle has corners $(2 - \mu, 2\mu - 2, 2 - \mu)$ and its permutations, side $\sqrt{2}\,(3\mu - 4)$, all against RD planes: $L_{tr} = 24\sqrt{2}\,(3\mu - 4)$, $L_{ts} = 0$. For $3/2 \le \mu \le 2$ the cube planes cut each triangle into a hexagon with corners such as $(1, \mu - 1, 2 - \mu)$: three sides $\sqrt{2}\,(2 - \mu)$ against RD planes and three sides $\sqrt{2}\,(2\mu - 3)$ against cube planes, so $L_{tr} - L_{ts} = 24\sqrt{2}\,[(2 - \mu) - (2\mu - 3)] = 24\sqrt{2}\,(5 - 3\mu)$. At $\mu = 2$ this is $-24\sqrt{2}$, the cuboctahedron's value in Theorem 1.1, as it must be. (b) Scale by 3: the planes become $|x_i| \le 3$, $|x_i| + |x_j| \le 5$, $|x| + |y| + |z| \le 6$, whose corners are the signed permutations of (1, 2, 3). (c) For $1 < \mu \le 4/3$ the dihedrals are $2\pi/3$ and $3\pi/4$ only. At a generic point of a $3\pi/4$ edge, $3\pi/4 \cdot x + 2\pi/3 \cdot y + \pi \cdot w =$

2π or π with $x \geq 1$; multiplying by $12/\pi$ gives $9x + 8y + 12w \in \{24, 12\}$, so x is even, $x = 2$, and $8y + 12w = 6$ or -6, neither possible. For the B3 permutohedron the dihedrals are $3\pi/4$ (rectangle–octagon), $\pi/2 + \beta$ (rectangle–hexagon) and $\pi - \beta$ (octagon–hexagon). At a generic point of a rectangle–octagon edge, $x(3\pi/4) + y(\pi/2 + \beta) + z(\pi - \beta) + w\pi \in \{2\pi, \pi\}$ with $x \geq 1$; the β-parts force $y = z$, and then $3x + 6y + 4w \in \{8, 4\}$, which again forces x even, $x = 2$, and $6y + 4w = 2$ or -2: no solution. The RD tiles space. ∎

The B3 permutohedron is the orbit of (3, 2, 1), the half-sum of positive roots of the group C3 (Sp(6)), so it is a weight shape in the sense of The Golden Ladder, Chapter 2 (DICTO DOI: 10.5281/zenodo.23966554). It has the same face directions and the same three dihedral angles as the Archimedean truncated cuboctahedron, with a different ratio of edges. Its hexagons are not centrally symmetric, so it is not a parallelohedron. Turned on its three-fold axis, it stays Dehn-zero and still fails the angle count; that halves-and-turns reading belongs to HexOct: The Hexagonality Law, Chapter 16 (DICTO DOI: 10.5281/zenodo.23731359), and is not repeated here.

4. The tetrahedron sweep

The tetrahedron's own dual sweep is $T \cap \mu(-T)$, from the tetrahedron ($\mu = 3$) through the Archimedean truncated tetrahedron ($\mu = 5/3$) to the octahedron ($\mu = 1$), with the mirror half $\mu \mapsto 1/\mu$ (#94, part E). For $1 \leq \mu \leq 3$ each corner is cut at $t = (3 - \mu)/4$ of the edge.

> **Proposition 1.3 (tetrahedron sweep).** For edge a, $D(T \cap \mu(-T)) = 6a\,(1 - 4t) \otimes \alpha$. It vanishes exactly at $t = 1/4$, $\mu = 2$ (and its mirror $\mu = 1/2$): the tetrahedron of edge 4 with its four corner tetrahedra of edge 1 removed. This solid tiles space alone.

Proof. The 6 hexagon–hexagon edges have length $(1 - 2t)a$ and dihedral α; the 12 hexagon–triangle edges have length ta and dihedral $\pi - \alpha$. So $D = 6(1 - 2t)a \otimes \alpha - 12ta \otimes \alpha$. The study checks the formula from the hull: coefficient 0 at $\mu = 2$, -5.657 at $\mu = 5/3$ ($= -6a(1/3)$ at the study's edge $a = 2\sqrt{2}$). The tiling is Theorem 2.4 below, case (1, 1, 1, 1). ∎

Known. The tiling by this truncated tetrahedron is known: Damasceno, Engel and Glotzer (2012) found it as the β-tin structure among the dense packings of a family of truncated tetrahedra, and Gabbrielli, Jiao and Torquato (2012, Figure 15) describe it by "chopping the four corners" at edge 1/4, each piece containing 20 tetrahedra and 10 octahedra of the octet. **New here:** that on all three sweeps the Dehn test alone picks out exactly the members that tile (with the angle count for the stretch and the B3 permutohedron of the CO–RD sweep); and the exact Dehn lines of Theorems 1.1 and 1.2.

Remark (the sweep's other measures). On the same tetrahedron sweep the roundness V^2/A^3 is largest exactly at the octahedron, and the Mahler volume is $4(\mu^3 - 9\mu^2 + 27\mu - 11)/(3\mu)$ for $1 \leq \mu \leq 3$, largest at $\mu^* = (11 - \sqrt{33})/4 \approx 1.3139$, where it is ≈ 11.3723 (study 2026-11-04-tetra-sweep, part 1). These are not space-filling results; their home is the roundness and Mahler thread of #94's next version.

5. Every corner cut

Let T be a regular tetrahedron of edge a and $c = (c_1, c_2, c_3, c_4)$ cut fractions, $c_i \geq 0$ and $c_i + c_j \leq 1$ (cuts may meet on an edge but not overlap). P_c is T with the corner tetrahedron of edge $c_i a$ removed at corner i.

> **Proposition 2.1.** $D(P_c) = 6a\,(1 - c_1 - c_2 - c_3 - c_4) \otimes \alpha$. So P_c is Dehn-zero exactly when $\Sigma c_i = 1$.

Proof. The edge between corners i and j keeps length $(1 - c_i - c_j)a$ and dihedral α. The cut triangle at corner i has three edges of length $c_i a$, each against a face through corner i, with dihedral $\pi - \alpha$ (the cut is parallel to the face opposite corner i). So $D = a[\Sigma_{i<j}(1 - c_i - c_j) - 3\Sigma c_i] \otimes \alpha = a(6 - 6\Sigma c_i) \otimes \alpha$, and $\alpha \notin \mathbb{Q}\pi$. ∎

The sweep of Section 4 is the uniform case $c_i = t$, which gives Proposition 1.3. The study states the family in a second form, which shows that it has one free parameter of position and none of shape.

> **Proposition 2.2 (one translation).** $\Sigma c_i = 1$ if and only if $P_c = T \cap (w + 2(-T))$ for some translation w: every Dehn-zero truncated tetrahedron is the tetrahedron cut by the inverted tetrahedron of twice its size, placed so that each of its four faces cuts off a corner of T or passes through it.

Proof. Write $T = \{x : -v_i \cdot x \le 1\}$, with $v_1, \ldots, v_4$ the corner directions, $\Sigma v_i = 0$ and $|v_i|^2 = 3$, so that corner i is v_i and $v_i \cdot x$ runs from -1 on the opposite face to 3 at corner i. The cut at corner i is the plane $v_i \cdot x \le \mu_i$ with $c_i = (3 - \mu_i)/4$. The four cut planes bound $\{v_i \cdot x \le \mu_i\} = w + m(-T)$, where $m = \Sigma\mu_i/4$ and w solves $v_i \cdot w = \mu_i - m$ (solvable, because the four equations add up to $0 = 0$ and any three v_i are independent). And $\Sigma c_i = (12 - \Sigma\mu_i)/4 = 1$ exactly when $\Sigma\mu_i = 8$, that is $m = 2$. ∎

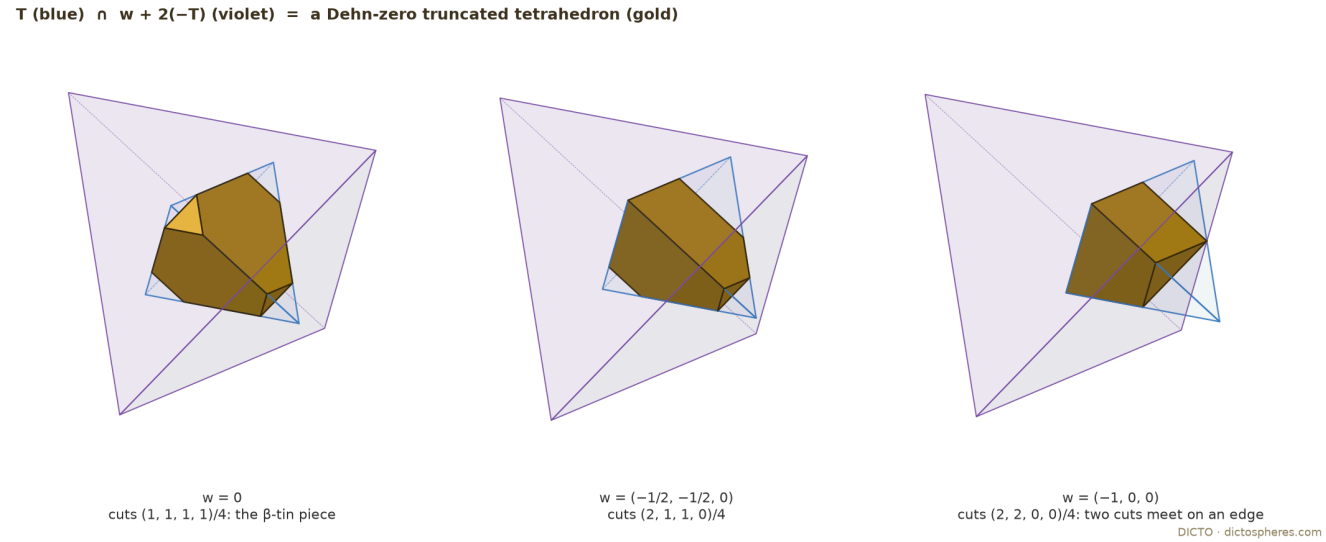


Figure 2. Proposition 2.2 at work. The tetrahedron T (blue, corners v_i = (1, 1, 1), (1, −1, −1), (−1, 1, −1), (−1, −1, 1), edge $2\sqrt{2}$) meets the inverted tetrahedron of twice its size, w + 2(−T) (violet), in a Dehn-zero truncated tetrahedron (gold). Left: w = 0 gives the uniform cuts 1/4, the β-tin piece. Middle: w = (−1/2, −1/2, 0) gives the cuts (2, 1, 1, 0)/4. Right: w = (−1, 0, 0) gives (2, 2, 0, 0)/4, whose two cuts meet on an edge. Moving the cutter changes the cuts but never their sum; the render script checks $\Sigma c_i = 1$ and the volume $(1 - \Sigma c_i^3)$ vol T for each panel.

6. Dehn zero is the octet's ratio

When a and the cuts are whole numbers, T(n) (edge n) is a union of the octet's cells of edge 1, and so is $P_k = T(n)$ with corner tetrahedra $T(k_i)$ removed, $\Sigma k_i \le n$, $k_i + k_j \le n$. The octet has exactly two tetrahedra for every octahedron, so a tiling by copies of an octet polyform that lies along the octet's cells needs that ratio in the piece.

> **Proposition 2.3 (density equals Dehn).** P_k has $n - \Sigma k_i$ more tetrahedra than twice its octahedra. So the octet ratio 2 : 1 holds in P_k exactly when $\Sigma k_i = n$, which is exactly when P_k is Dehn-zero.

Proof. T(m) consists of C(m + 2, 3) upward and C(m, 3) downward tetrahedra and C(m + 1, 3) octahedra, so its excess (tetrahedra minus twice the octahedra) is the second difference C(m + 2, 3) − 2C(m + 1, 3) + C(m, 3) = m. Excess is additive, so P_k has excess $n - \Sigma k_i$. With $c_i = k_i/n$, Proposition 2.1 finishes. ∎

For the uniform cut (1, 1, 1, 1) the study noted this at n = 4: T(4) minus its corners is 20 tetrahedra and 10 octahedra, the counts of Gabbrielli, Jiao and Torquato. Proposition 2.3 is the same remark for every cut: the global test (Dehn) and the counting test (density) coincide on the whole family.

7. Every case tested tiles

The Dehn-zero integer pieces with n ≤ 5 are the twelve partitions of n into two to four parts: (1, 1) at n = 2; (2, 1), (1, 1, 1) at n = 3; (3, 1), (2, 2), (2, 1, 1), (1, 1, 1, 1) at n = 4; (4, 1), (3, 2), (3, 1, 1), (2, 2, 1), (2, 1, 1, 1) at n = 5 (zeros omitted).

> **Theorem 2.4 (computer-checked, exact cover).** Each of the twelve Dehn-zero integer truncations P_k with n ≤ 5, and $P_{(5,3,0,0)}$ at n = 8, tiles space periodically along the octet's cells. In every case the smallest period found holds two pieces, the piece and its point inverse; no torus with one piece per period is tiled.

Proof (computation). For each piece, every orientation in the octet's symmetry group, and every period lattice of index N in the face-centred cubic lattice of the octet's corners (all Hermite normal forms, each fcc point carrying one octahedron and two tetrahedra), the exact cover problem "each cell of the torus in exactly one copy" is solved by the SAT solver CaDiCaL, starting at N = number of octahedra in the piece (one piece per period) and then at twice that. Every case has no solution at one piece per period and a solution at two (Table 2.1). The cells are exact, so a solution is a tiling, not an approximation. The method was calibrated on the two pieces tiled earlier by the exact-cover search, (1, 1, 1, 1) at N = 20 and (1, 1, 0, 0) at N = 2. As a second, geometric check, the (1, 1, 1, 1) tiling at N = 20 was built with real polytopes: 2 × 20 = 40 = the period's volume, no overlap among 18 489 nearby pairs, and every one of 15 512 random points within radius 5.9 in exactly one copy; the pieces (2, 1, 0, 0)/3 and (1, 1, 1, 0)/3 passed the same check in their mirror-pair form (0 overlaps, 4000 of 4000 points covered once), and a deliberately wrong offset failed it (213 overlaps). ∎

n	cuts k	octahedra	tetrahedra	N (two pieces)	period (HNF)
2	(1, 1, 0, 0)	1	2	2	diag(1, 1, 2)
3	(2, 1, 0, 0)	3	6	6	diag(1, 2, 3)
3	(1, 1, 1, 0)	4	8	8	diag(2, 2, 2)
4	(3, 1, 0, 0)	6	12	12	diag(1, 3, 4)
4	(2, 2, 0, 0)	8	16	16	diag(2, 2, 4)
4	(2, 1, 1, 0)	9	18	18	diag(2, 3, 3)
4	(1, 1, 1, 1)	10	20	20	[[10, 1, 4], [0, 2, 1], [0, 0, 1]]
5	(4, 1, 0, 0)	10	20	20	diag(1, 4, 5)
5	(3, 2, 0, 0)	15	30	30	diag(2, 3, 5)
5	(3, 1, 1, 0)	16	32	32	diag(2, 4, 4)
5	(2, 2, 1, 0)	18	36	36	diag(3, 3, 4)
5	(2, 1, 1, 1)	19	38	38	[[19, 3, 9], [0, 1, 0], [0, 0, 2]]
8	(5, 3, 0, 0)	60	120	120	diag(3, 5, 8)

Table 2.1. The thirteen tilings. N is the index of the period lattice in the fcc lattice; N equals twice the number of octahedra, so each period holds two pieces. The patches of radius 7.2 to 12 close for every n ≤ 5 piece as well (SAT patch test). Data: studies/2026-11-04-tetra-sweep, census-sat.log, satper-1.log, satper-2.log (study data), census-n4.log, tiling-N20.json, verify-real.log.

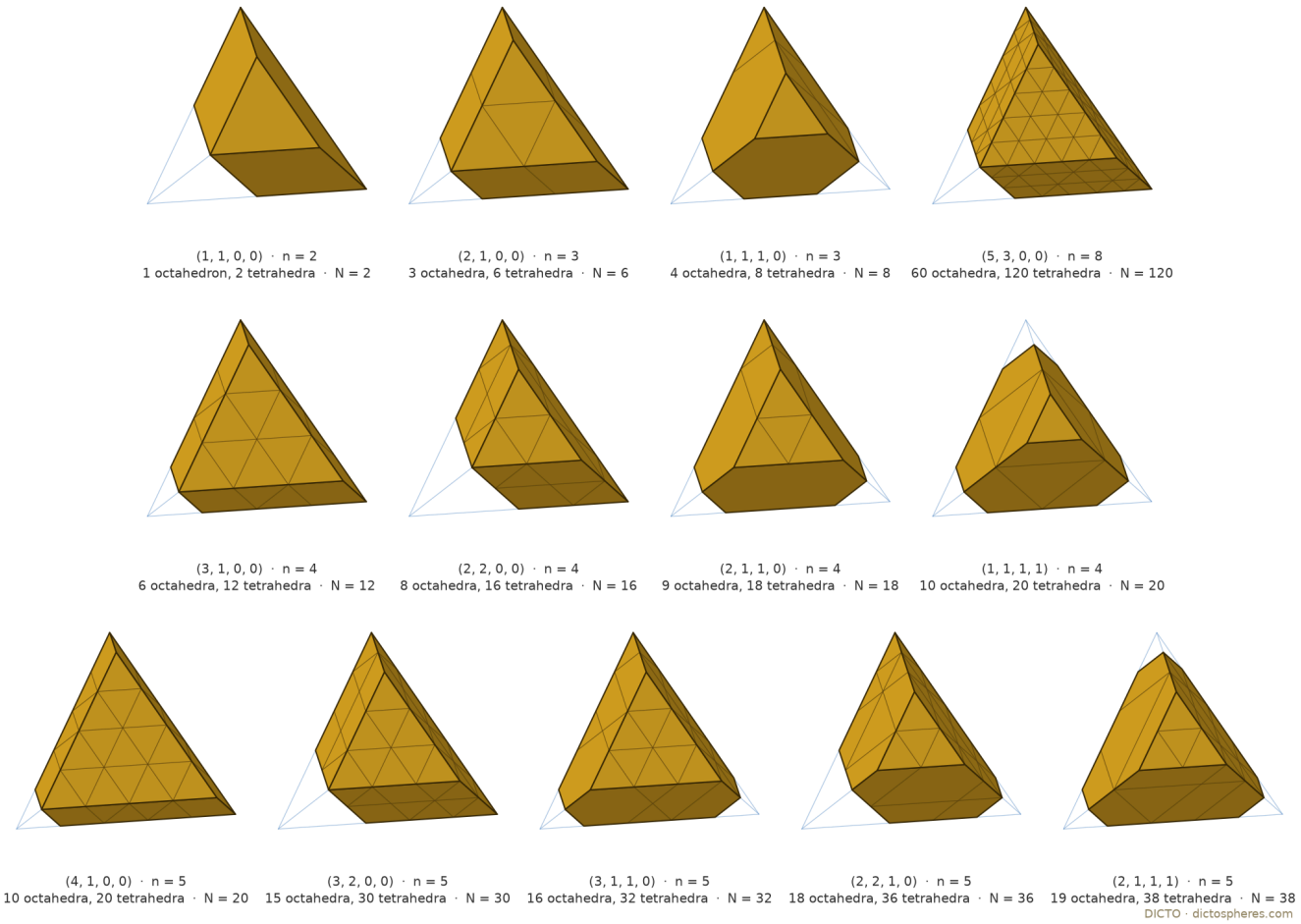


Figure 3. The thirteen pieces of Theorem 2.4, each drawn inside its tetrahedron T(n) (blue outline) and scaled to the same size; the thin lines on the faces are the octet's planes, so the triangles of edge 1 that build each piece show. Under each: the cuts k, the octahedra and tetrahedra in the piece (always one to two, Proposition 2.3), and the index N of the smallest period found, which holds two pieces. The cuts are drawn on the corners nearest the viewer, largest first. (1, 1, 0, 0) at n = 2 and (2, 2, 0, 0) at n = 4 are the same solid at two sizes against the octet. The render script recounts the cells of every piece and checks them against Table 2.1.

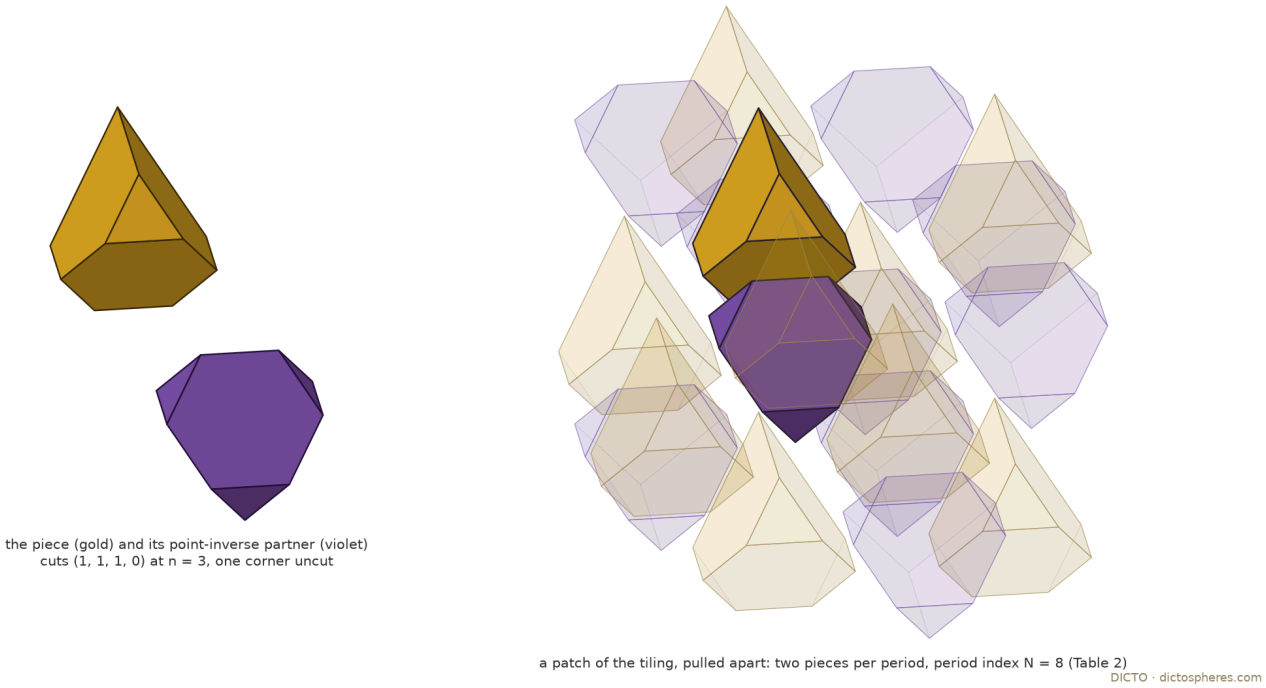


Figure 4. One of the new tilings: the piece with cuts (1, 1, 1, 0) at n = 3, one corner left sharp. Left: the piece (gold) and its partner (violet), the mirror image $\sigma_z P$ shifted by $w = b_1/3$, which is a turned copy of the piece's point inverse. Right: a patch of the tiling pulled apart from its centre; one piece and its partner solid, their neighbours as glass. Period: index N = 8 (Table 2.1). Built exactly as in the study's real-polytope check (scripts/verify_real.py; verify-real.log: 0 overlaps among 31 138 pairs, 4000 of 4000 points covered once); the render script checks again that all of 6000 random points lie in exactly one copy.

> **Conjecture 2.5.** Every Dehn-zero truncated tetrahedron with rational cuts tiles space, as the piece together with its point inverse, two to a period.

For irrational cuts it stays open, and the first construction tried does not settle it. The study tried one uniform mirror-pair rule, P together with its mirror image $\sigma_z P$ shifted by w on the lattice $B \cdot \mathrm{diag}(1 - c_1, 1 - c_2, 1 - c_3)$, which works for the integer cuts at their own offsets. For the golden cut $(\varphi^{-1}, \varphi^{-2}, 0, 0)$ at $w = 0$ it fails (436 overlapping pairs of 41 412; 388 of 4000 points uncovered). The scan over other offsets has no recorded result. So the conjecture is stated for rational cuts only. A proof would need one construction that works continuously in the translation w of Proposition 2.2; the solver's periods (Table 2.1) do not yet show one.

8. The golden members

The next pieces to try are those with golden cuts. This section gives only the finished part of that study, the exact list.

> **Proposition 2.6 (golden Dehn-zero cuts).** Among cut vectors with every c_i in $\{0, \varphi^{-1}, \varphi^{-2}, \ldots, \varphi^{-8}\}$ and at least two cuts nonzero, those with $\Sigma c_i = 1$ and $c_i + c_j \le 1$ are exactly the seven of Table 2.2, up to the order of the corners.

Proof (exact computation). Every combination with repetition of four values from the list is summed and compared in $\mathbb{Z}[\varphi]$, written $a + b\varphi$ with $\varphi^2 = \varphi + 1$, so no floating point is used; the condition $c_i + c_j \le 1$ is decided exactly by comparing $b\sqrt{5}$ with an integer. Each sum can be checked by hand from $\varphi^{-k} + \varphi^{-k-1} = \varphi^{-k+1}$. ∎

#	cuts	Fibonacci approximants n: k	approximants that tile (Table 2.1)
1	0, 0, φ^{-1}, φ^{-2}	3: (0, 0, 2, 1) · 5: (0, 0, 3, 2) · 8: (0, 0, 5, 3) · 13: (0, 0, 8, 5)	n = 3, 5, 8
2	0, φ^{-1}, φ^{-3}, φ^{-4}	3: (0, 2, 1, 0) · 5: (0, 3, 1, 1) · 8: (0, 5, 2, 1) · 13: (0, 8, 3, 2)	n = 3, 5
3	0, φ^{-2}, φ^{-2}, φ^{-3}	3: (0, 1, 1, 1) · 5: (0, 2, 2, 1) · 8: (0, 3, 3, 2) · 13: (0, 5, 5, 3)	n = 3, 5
4	φ^{-1}, φ^{-3}, φ^{-5}, φ^{-6}	8: (5, 2, 1, 0) · 13: (8, 3, 1, 1) · 21: (13, 5, 2, 1) · 34: (21, 8, 3, 2)	none tested (first at n = 8)
5	φ^{-1}, φ^{-4}, φ^{-4}, φ^{-5}	5: (3, 1, 1, 0) · 8: (5, 1, 1, 1) · 13: (8, 2, 2, 1) · 21: (13, 3, 3, 2)	n = 5
6	φ^{-2}, φ^{-2}, φ^{-4}, φ^{-5}	5: (2, 2, 1, 0) · 8: (3, 3, 1, 1) · 13: (5, 5, 2, 1) · 21: (8, 8, 3, 2)	n = 5
7	φ^{-2}, φ^{-3}, φ^{-3}, φ^{-4}	3: (1, 1, 1, 0) · 5: (2, 1, 1, 1) · 8: (3, 2, 2, 1) · 13: (5, 3, 3, 2)	n = 3, 5

Table 2.2. The seven golden Dehn-zero truncated tetrahedra, with Fibonacci approximants ($\varphi^{-j} \approx F_{m-j}/F_m$, chosen so that the k add up exactly to $n = F_m$). Piece 1 has 11 corners, because its two cuts meet on an edge ($\varphi^{-1} + \varphi^{-2} = 1$); the others have 12. Data: studies/2026-11-04-tetra-sweep/data/golden-cuts.json, scripts/golden_cuts.py.

Six of the seven have small approximants that tile, and piece 1 has three in a row. That is evidence, not a result: whether a golden piece itself tiles is not decided. The earlier patch searches on these approximants timed out (census-golden.log) before the SAT method replaced them, and the one construction tried on the real golden cut fails (Section 7). None of the seven is harmonically silent: in #94's fingerprint each has Q_3, Q_4, Q_6, Q_8 all nonzero (golden-harmonic.json), so the harmonic test gives no further clue.

9. What is known, and what is new

Known, credited: the Dehn invariant (Dehn 1901), Debrunner's theorem (1980), Sydler's theorem (1965); the tilings by Kelvin's cell, the cube and the RD; the octet (Lagarias and Zong 2012); the uniform member T(4) minus its corners and its β-tin tiling (Damasceno, Engel and Glotzer 2012; Gabbrielli, Jiao and Torquato 2012, Fig. 15, with its counts 20 and 10). **DICTO's earlier work, cited:** the sweeps (#90, #94); the weight shapes (The Golden Ladder, Chapter 2). **New here:** Theorems 1.1 and 1.2 with their exact Dehn lines; the identification of the CO–RD sweep's lone Dehn-zero solid as the B3 permutohedron and the proof that it cannot tile alone; that the cube-trimmed RDs are Dehn-zero but cannot tile alone; that on all three sweeps the Dehn test, with the angle count, picks out exactly the members that tile; Propositions 2.1 to 2.3 (the Dehn-zero family by its cut sum, by one translation of 2(−T), and by the octet ratio); the twelve tilings of Theorem 2.4 beyond the uniform one, and the thirteenth at n = 8; Conjecture 2.5; the list of seven golden pieces. In the sources checked, only the uniform cut 1/4 appears.

Sources checked: the studies' own tables (standard dihedral tables of the Platonic and Catalan solids, 2026-11-04-rt-pyramids); the reference library entries Damasceno, Engel and Glotzer, ACS Nano 6 (2012) 609, and Gabbrielli, Jiao and Torquato, Phys. Rev. E 86 (2012) 041141, both read for the β-tin tiling (2026-11-04-tetra-sweep, part 2); Conway, Radin and Sadun (1999), in the library, for the irrationality of the angles. No search was

made for earlier statements of Theorems 1.1 and 1.2 beyond these, and no wider search yet for non-uniform truncated tetrahedra that tile (status box).

10. Data, scripts and checks

All in the research branch; heavy runs as logged jobs. **Sweeps:** studies/2026-11-04-rt-pyramids (README, section "Space-filling (Dehn) test on the sweeps' own solids"); studies/2026-11-04-tetra-sweep (scripts/measures.py, data/measures.json); the Concordance grid, row "CO–RD sweep", column "Dehn", corrected 4 November 2026 from "zero only at μ = 5/3" to the stretch 1 ≤ μ ≤ 4/3 as well. **Dehn screen** (used to find the statements, not to prove them; every Dehn statement here is proved exactly): studies/2026-11-04-dehn-screen, scripts/dehn.py, calibrated 11 of 11 at tolerance 10^{-8}. **Truncated tetrahedra:** studies/2026-11-04-tetra-sweep, README parts 1, 2 and "Reformulation"; scripts census.py, sat_periodic.py (exact cover with CaDiCaL through PySAT; calibrated on the 60° rhombohedron, which tiles at index 1, a lone tetrahedron, which never tiles, and the two earlier tilings), verify_tiling.py, verify_real.py, golden_cuts.py, golden_harmonic.py; data census-sat.log, satper-1.log, satper-2.log, census-n4.log, tiling-N20.json, verify-real.log, golden-cuts.json, golden-harmonic.json. A first enumeration of the period lattices used wrong off-diagonal ranges (right count, wrong lattices) and missed N = 20; it was fixed before any result here. Concordance ledger, entries "Census", "SAT periodic test", "Periodic test", "CONJECTURE", "Uniform mirror-pair construction on REAL golden cuts". **Figures:** scripts/figlib.py (a small orthographic renderer) with fig_sweeps.py, fig_cutter.py, fig_table.py, fig_tiling.py in studies/2026-11-04-tetra-sweep/scripts. Each re-checks what it draws: fig_sweeps.py recomputes the Dehn coefficient of all 18 drawn solids from their hulls and matches the formulas of Theorems 1.1, 1.2 and Proposition 1.3; fig_cutter.py checks $\Sigma c_i = 1$ and the volume for each cutter position; fig_table.py recounts the octahedra and tetrahedra of all thirteen pieces against Table 2.1 and the piece volumes; fig_tiling.py rebuilds the (1, 1, 1, 0)/3 tiling and finds all of 6000 random points in exactly one copy.

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